

Benchmarking Edge Deployment of DRL Navigation Policies: Latency, Compression, and Real-Robot Validation

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6.3×

faster runtime

ONNX Runtime vs. baseline

71%

smaller model

static INT8 artifact

72.0%

paired success

INT8; 200 matched trials

10/10

real-robot arrivals

center block; no contact

Benchmark deployment cost and navigation behavior separately: faster inference does not by itself imply safer or more reliable navigation.

WHY A BENCHMARK?

Edge deployment changes runtime, storage, and closed-loop behavior. Reporting only one number can hide a trade-off elsewhere. We therefore evaluate execution efficiency, paired navigation outcomes, deployment optimization, and physical transfer under one controlled protocol.

KEY QUESTION

Which deployment changes reduce execution cost, and which navigation behaviors remain unchanged?

BENCHMARK DESIGN

01

Configure

policy × model form × hardware

02

Measure

latency, tails, size

03

Pair

identical simulated scenarios

04

Validate

arrival, contact, shield

SCOPE

999 paired general scenarios
200 paired compression trials
50 formal physical trials
CNNTD3, SAC, PPO, and NeuPAN

Controls: warm batch-one timing;
explicit GPU synchronization;
restored paired scenes; exact
McNemar tests; Wilson intervals.

WHAT THE BENCHMARK REVEALS

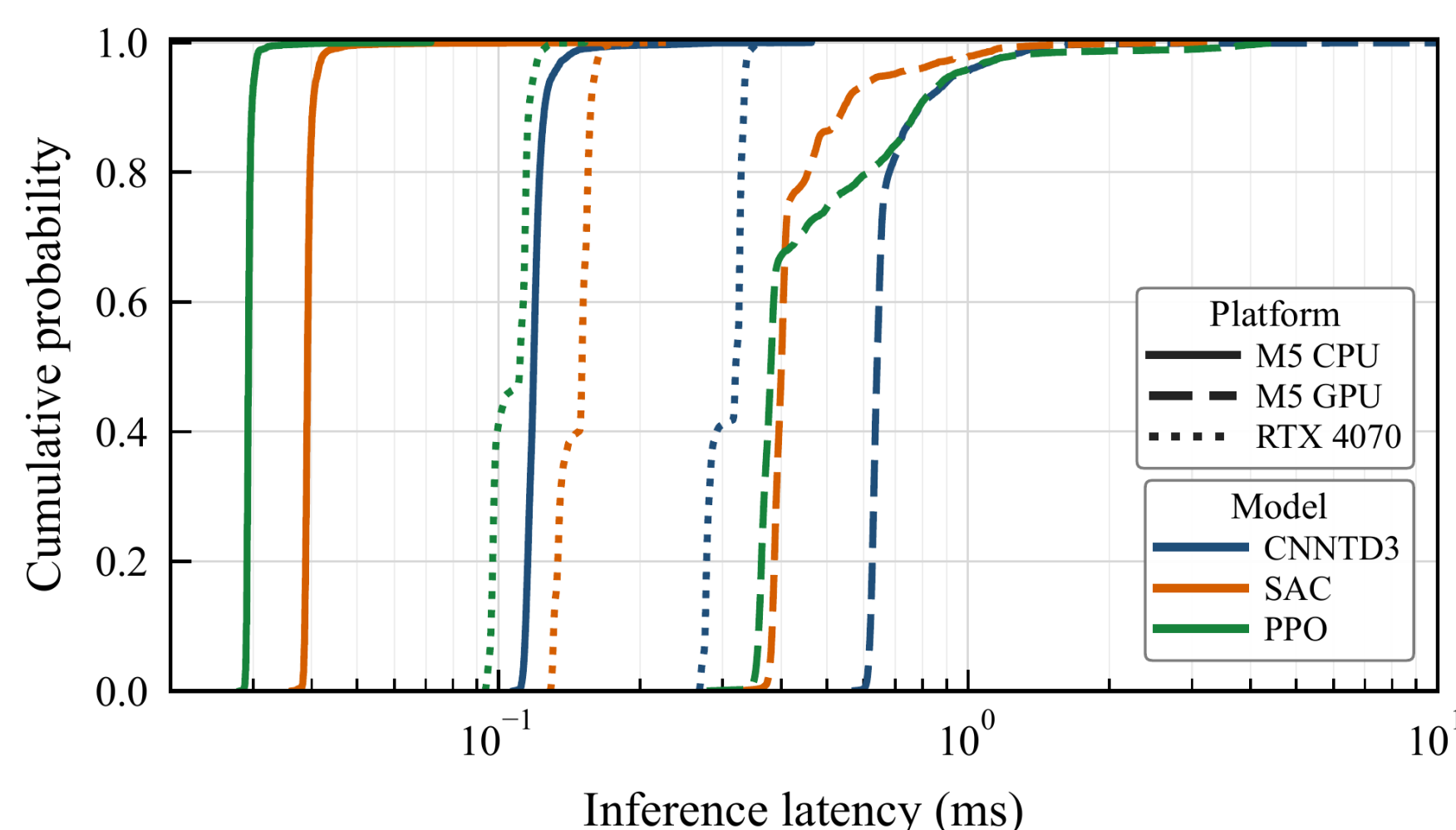
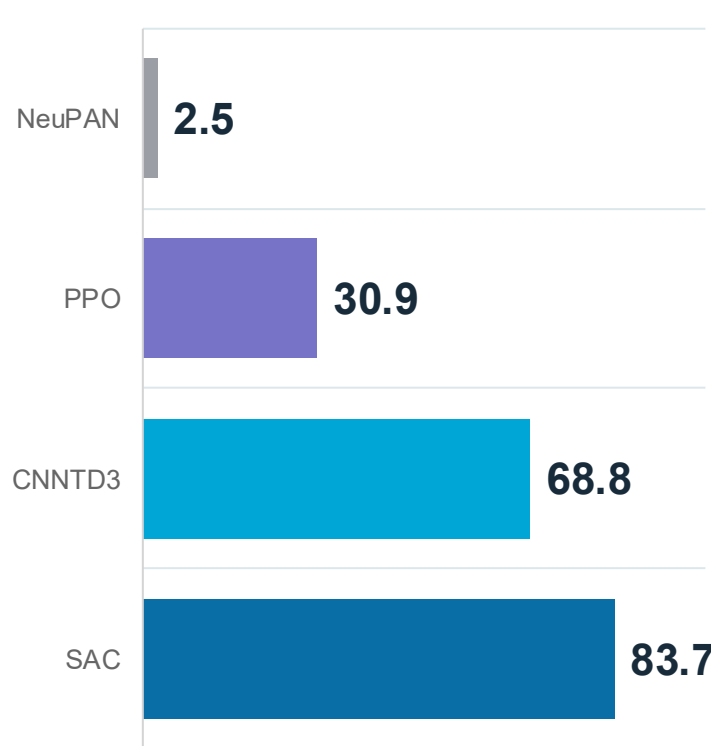


Fig. 1. Model-only latency distributions. DRL means span 0.03–0.70 ms off robot; CPU timing is the most concentrated on the tested stacks.

Finding 1 • Low latency and navigation success are orthogonal.



General benchmark success (%)

Version	Size	Latency	Succ.
Baseline	0.538	0.1204	72.5%
Pruned 50%	0.164	0.1221	61.0%
Pruned+FT	0.164	0.1177	70.0%
ORT FP32	0.524	0.0191	72.0%
ORT INT8	0.157	0.0282	72.0%

Size (MiB), model-only latency (ms), paired success

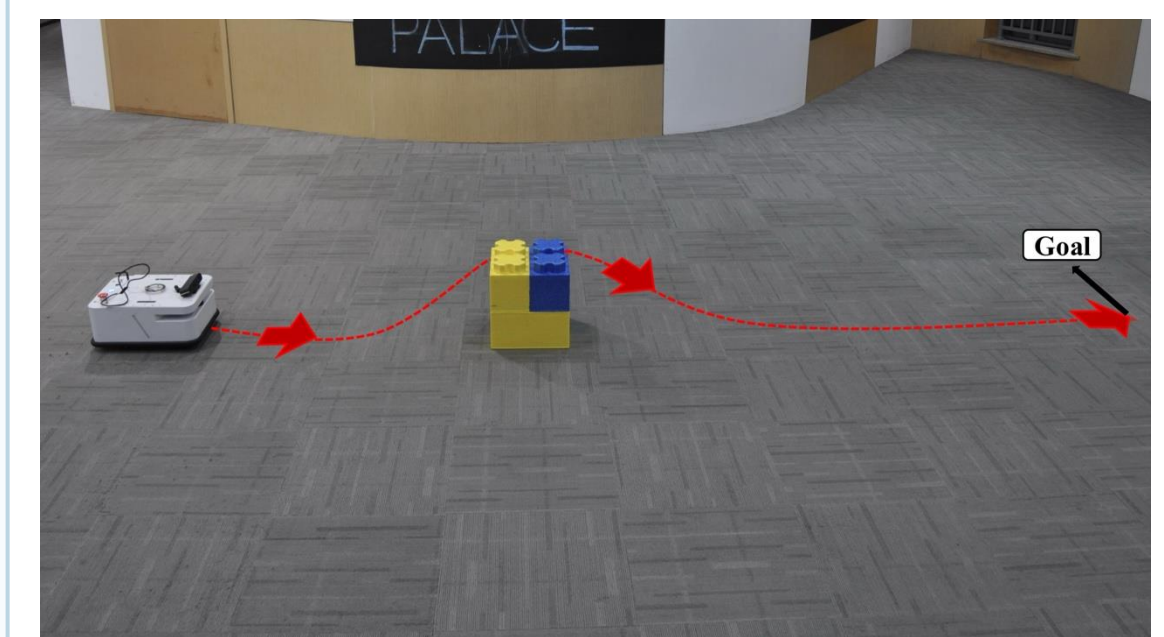
Finding 2 • Runtime migration gives the main speedup; INT8 gives the main size reduction.

ONNX Runtime reduced mean latency from 0.1204 to 0.0191 ms (6.3×). Static INT8 reduced the artifact from 0.538 to 0.157 MiB (71%) while retaining 72.0% success in the same 200 paired scenarios.

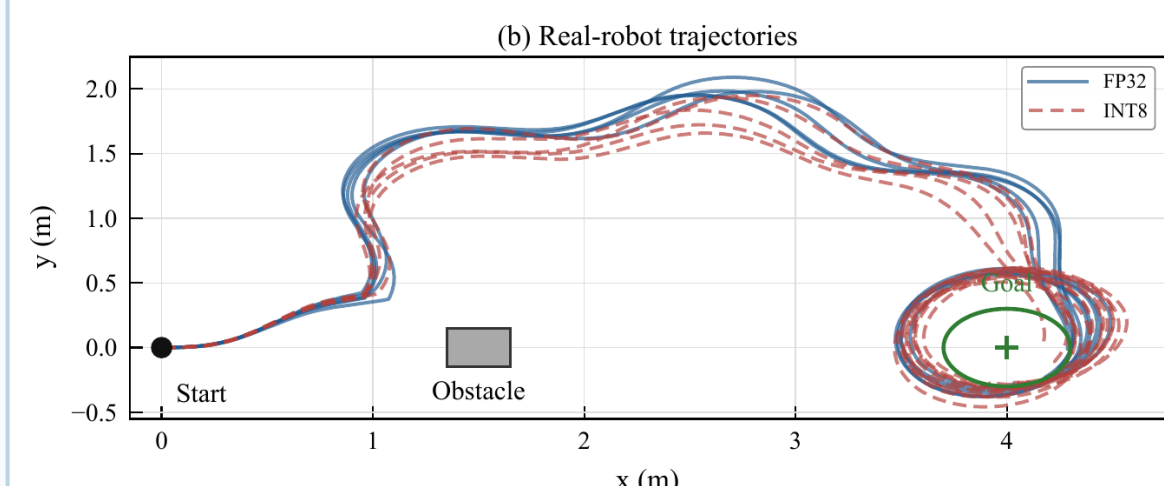
NeuPAN averaged 10.255 ms but succeeded in only 2.5% of trials under the evaluated interface and environment setup, reinforcing that timing and task behavior must be reported separately.

Model-only inference is only one part of deployment. On the robot, the measured sensing-to-command path was 87–89 ms.

WHAT TRANSFERRED TO THE ROBOT



(a) Center-blocking experimental setup.



(b) Five FP32 and five INT8 formal trajectories.

CENTER-BLOCKING TRANSFER

FP32 5/5 contact-free
INT8 5/5 contact-free
Matched C1 simulation 20/20

Across formal trials:

- straight: 10/10 arrivals
- center block: 10/10 arrivals
- side obstacles: 20/20 arrivals, with four left-side contacts

OBSERVED BOUNDARY

Low-clearance path overlap: 2/10 arrivals, 2/10 contacts, 8/10 shield activations, and 0/10 composite successes. This is a transfer boundary, not evidence of general navigation reliability.

TAKEAWAY • Runtime, storage, task behavior, and physical evidence answer different deployment questions—report all four.